Group 1

Lab 12 – Ethics, Fairness, and Bias in ML

ITAI-1371

Viswanatha Rao

Team Member Contribution Journal

Viktoriya Kurmisheva:

Worked on Task 2, Deeper Dive with a Confusion Matrix, which focused on calculating and comparing the False Positive Rate (FPR) and False Negative Rate (FNR) for males and females. Used the confusion matrix results to examine how often the model made different types of prediction errors for each group. Answered Question 2 in the Reflective Knowledge Check by interpreting these differences and explaining how incorrect income predictions could affect loan approval decisions. After completing the lab, took responsibility for writing and organizing the group's Reflection Journal based on the completed tasks, results, and discussions about fairness and bias in machine learning. Prepared the journal for upload to the group's GitHub repository.

Tien Manh Nguyen:

Worked on Task 1 in Part 2, which involved calculating the model's prediction accuracy separately for males and females using the Adult Income dataset. Compared the results to determine whether the model performed better for one group and examined the differences in accuracy. Answered Question 1 in the Reflective Knowledge Check by analyzing these results and explaining why high overall accuracy does not necessarily mean a model performs equally well for everyone. After the group completed the assigned tasks, took responsibility for reviewing the notebook to check that the code, calculations, and written answers matched the lab requirements. Prepared the completed Jupyter Notebook for upload to the group's GitHub repository.

Zaid Alayedy:

Worked on Questions 3 and 4 in the Reflective Knowledge Check, which focused on the ethical consequences of using machine learning models and possible ways to reduce bias. For Question 3, analyzed how False Positive and False Negative errors could affect hiring decisions and explained why the model should not be used for hiring without further evaluation. For Question 4, examined why removing the sex column would not completely eliminate bias, since other features might still contain information related to gender. Discussed possible improvements, including reweighting and fairness constraints. Reviewed the initial model setup and its connection to the ethical concerns explored in the lab. Took responsibility for gathering each member's contribution details and organizing the Team Member Contribution Journal for submission.